

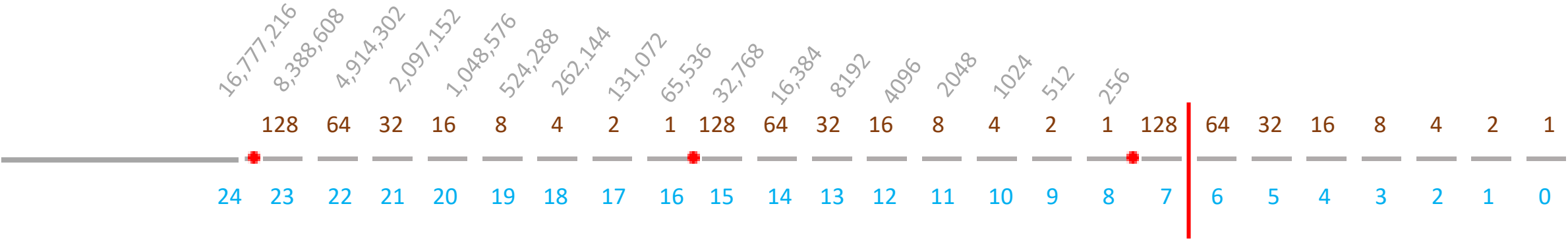
Subnetting by Host Requirements

Task 1

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

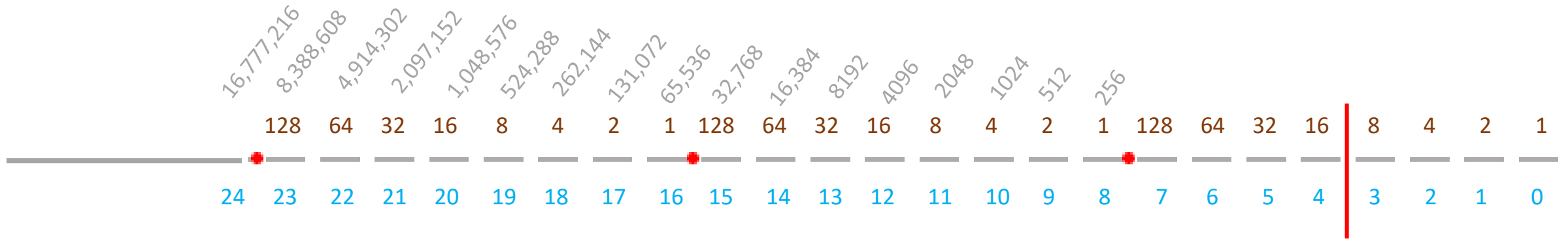
$2^n - 2 = \text{hosts}$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
126	7	128	4 th	/25	255.255.255.128

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

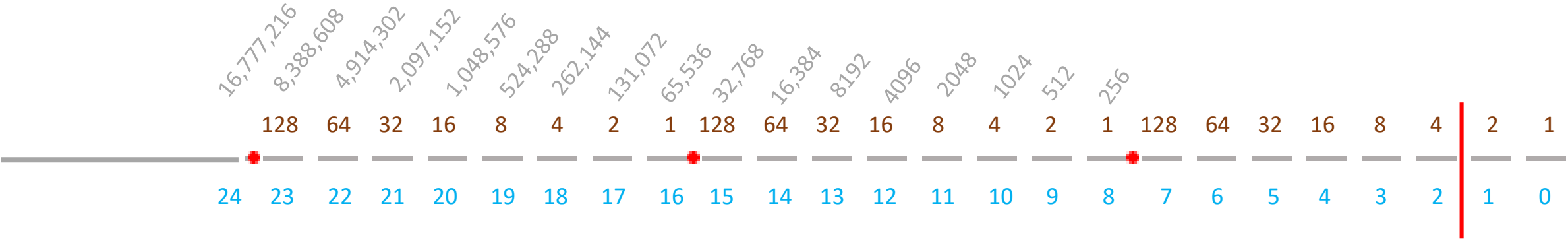
$$2^n - 2 = \text{hosts}$$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
126	7	128	4 th	/25	255.255.255.128
14	4	16	4 th	/28	255.255.255.240

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

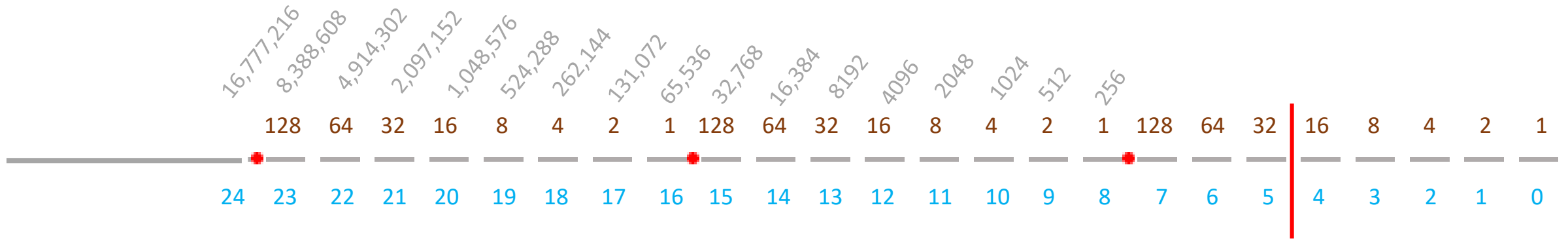
2^n - 2 = hosts



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
126	7	128	4 th	/25	255.255.255.128
14	4	16	4 th	/28	255.255.255.240
2	2	4	4 th	/30	255.255.255.252

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

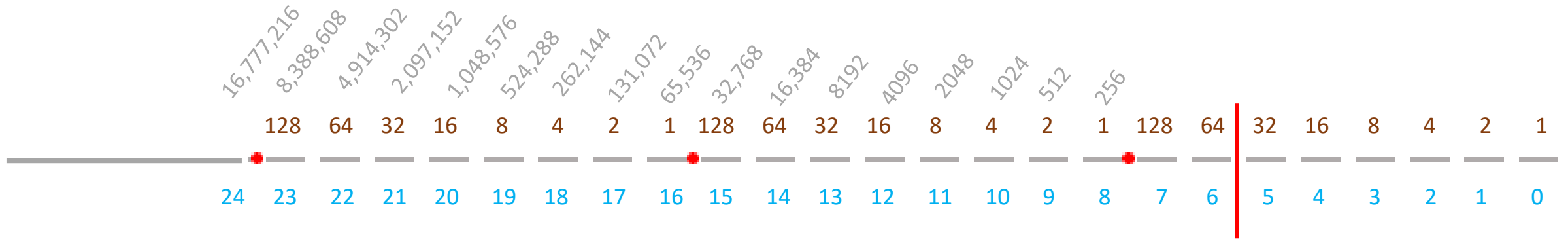
$$2^n - 2 = \text{hosts}$$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
126	7	128	4 th	/25	255.255.255.128
14	4	16	4 th	/28	255.255.255.240
2	2	4	4 th	/30	255.255.255.252
30	5	32	4 th	/27	255.255.255.224

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

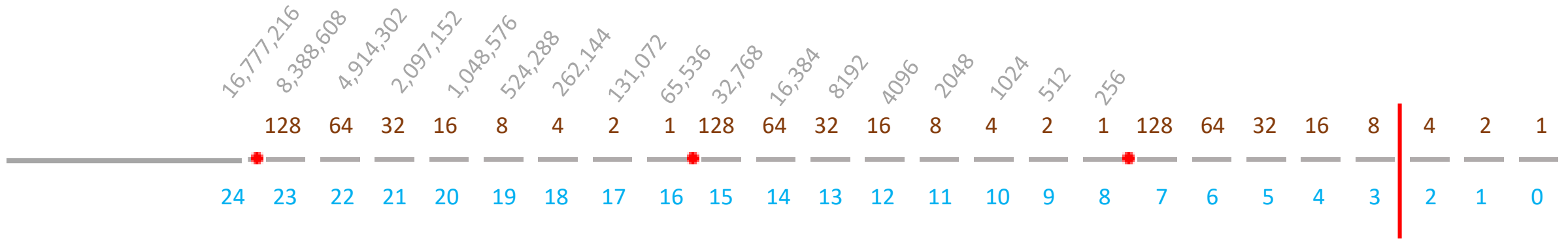
$$2^n - 2 = \text{hosts}$$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
126	7	128	4 th	/25	255.255.255.128
14	4	16	4 th	/28	255.255.255.240
2	2	4	4 th	/30	255.255.255.252
30	5	32	4 th	/27	255.255.255.224
62	6	64	4 th	/26	255.255.255.192

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

$$2^n - 2 = \text{hosts}$$



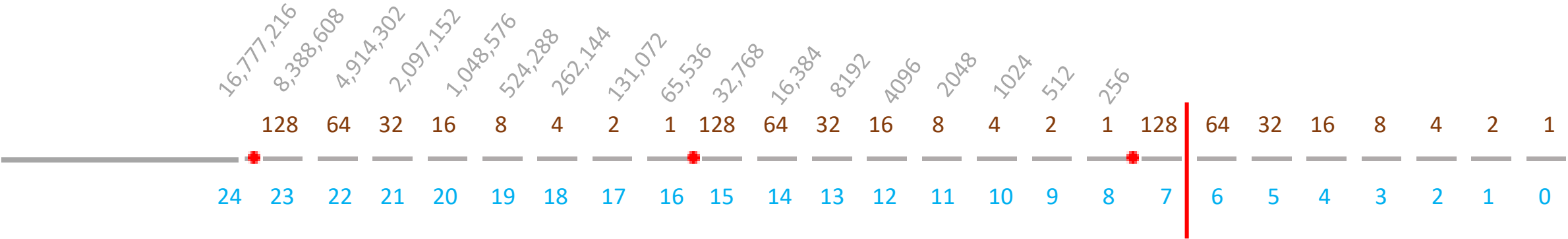
Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
126	7	128	4 th	/25	255.255.255.128
14	4	16	4 th	/28	255.255.255.240
2	2	4	4 th	/30	255.255.255.252
30	5	32	4 th	/27	255.255.255.224
62	6	64	4 th	/26	255.255.255.192
6	3	8	4 th	/29	255.255.255.248

Task 2

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

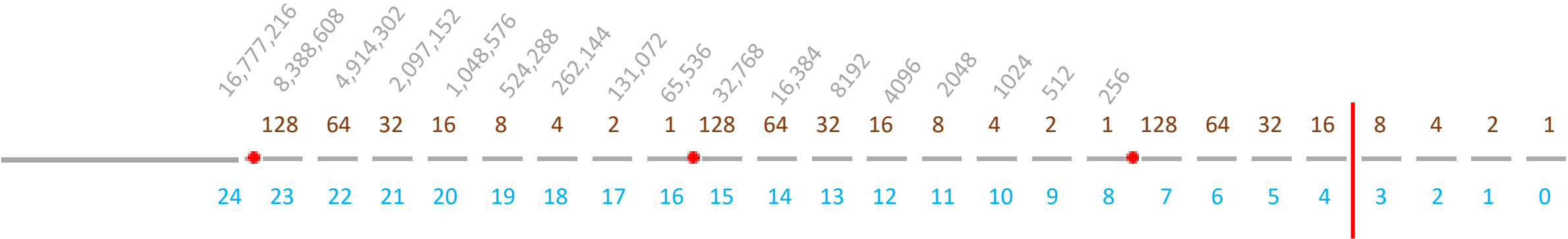
$2^n - 2 = \text{hosts}$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
64	7	128	4 th	/25	255.255.255.128

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

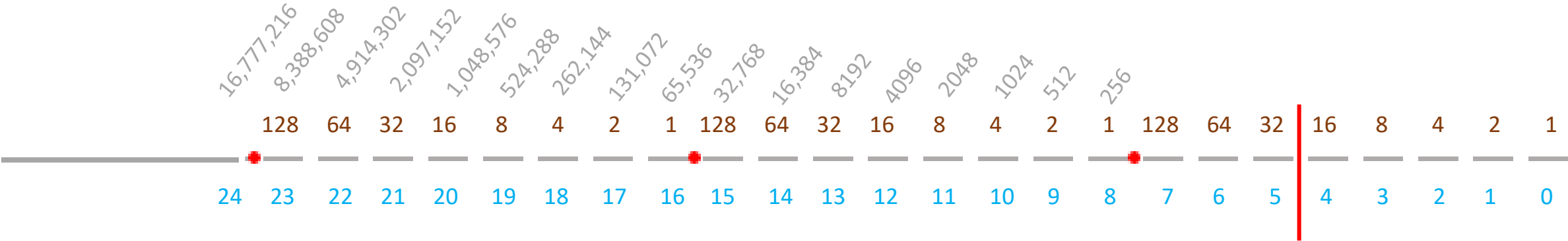
2^n - 2 = hosts



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
64	7	128	4 th	/25	255.255.255.128
7	4	16	4 th	/28	255.255.255.240

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

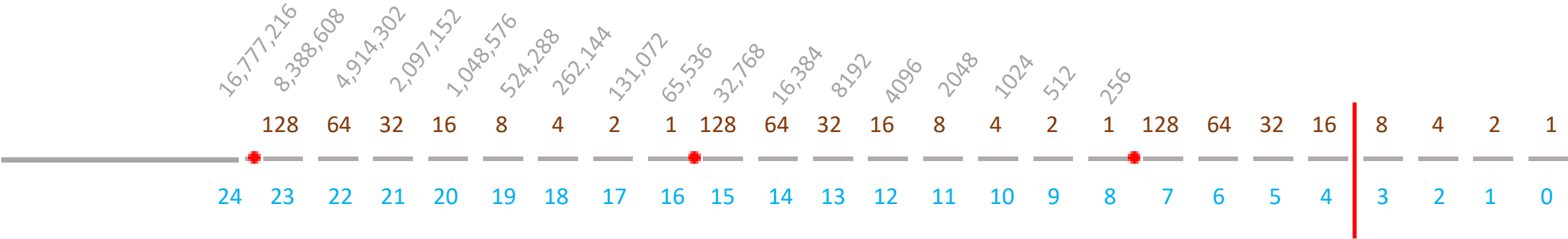
2^n - 2 = hosts



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
64	7	128	4 th	/25	255.255.255.128
7	4	16	4 th	/28	255.255.255.240
15	5	32	4 th	/27	255.255.255.224

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

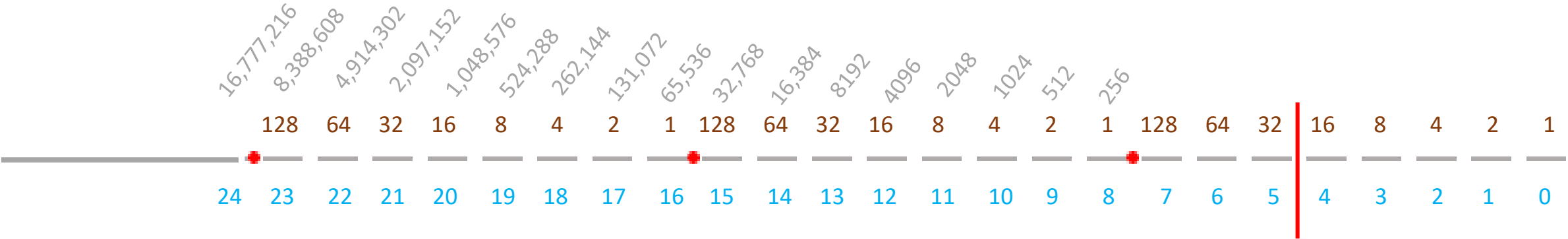
2^n - 2 = hosts



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
64	7	128	4 th	/25	255.255.255.128
7	4	16	4 th	/28	255.255.255.240
15	5	32	4 th	/27	255.255.255.224
10	4	16	4 th	/28	255.255.255.240

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

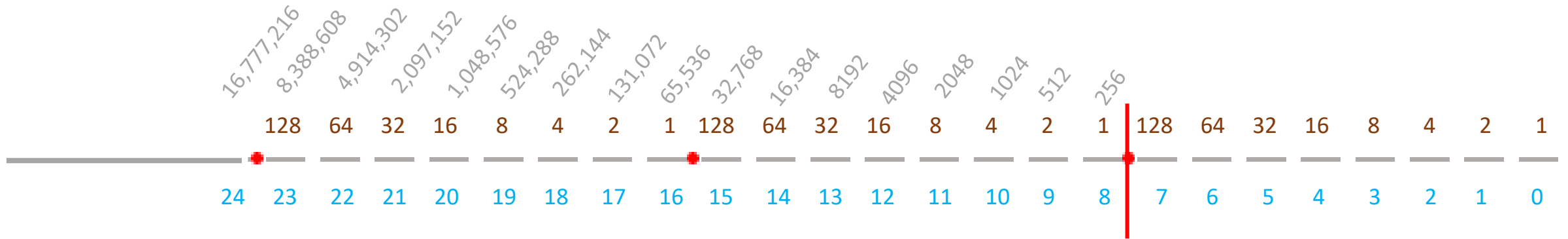
2^n - 2 = hosts



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
64	7	128	4 th	/25	255.255.255.128
7	4	16	4 th	/28	255.255.255.240
15	5	32	4 th	/27	255.255.255.224
10	4	16	4 th	/28	255.255.255.240
28	5	32	4 th	/27	255.255.255.224

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

$$2^n - 2 = \text{hosts}$$



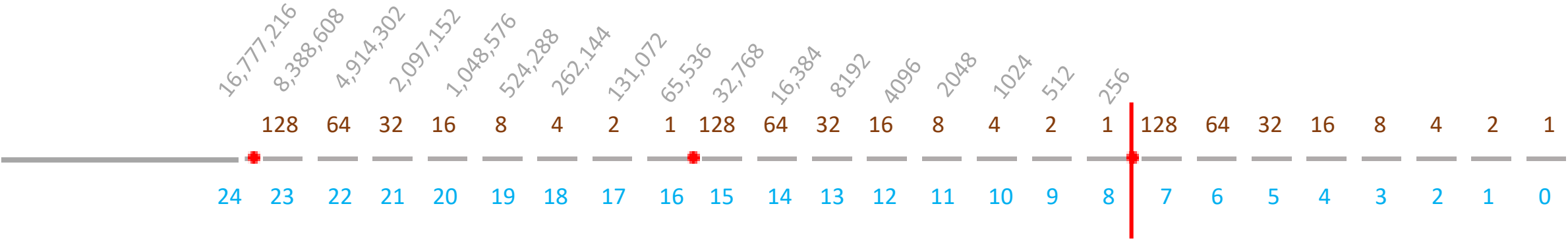
Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
64	7	128	4 th	/25	255.255.255.128
7	4	16	4 th	/28	255.255.255.240
15	5	32	4 th	/27	255.255.255.224
10	4	16	4 th	/28	255.255.255.240
28	5	32	4 th	/27	255.255.255.224
200	8	1	3 rd	/24	255.255.255.0

Task 3

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

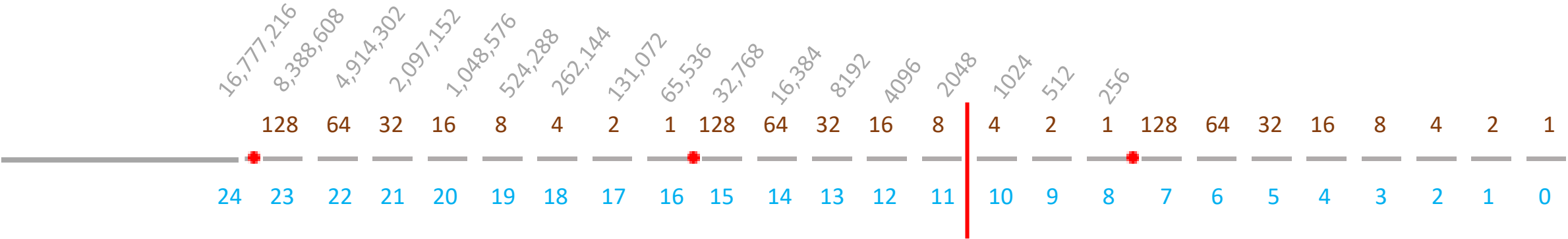
$2^n - 2 = \text{hosts}$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
250	8	1	3 rd	/24	255.255.255.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

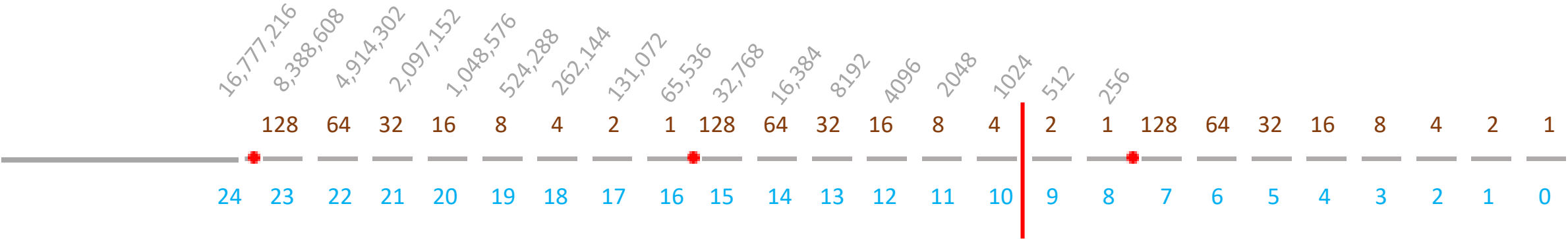
2^n - 2 = hosts



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
250	8	1	3 rd	/24	255.255.255.0
2000	11	8	3 rd	/21	255.255.248.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

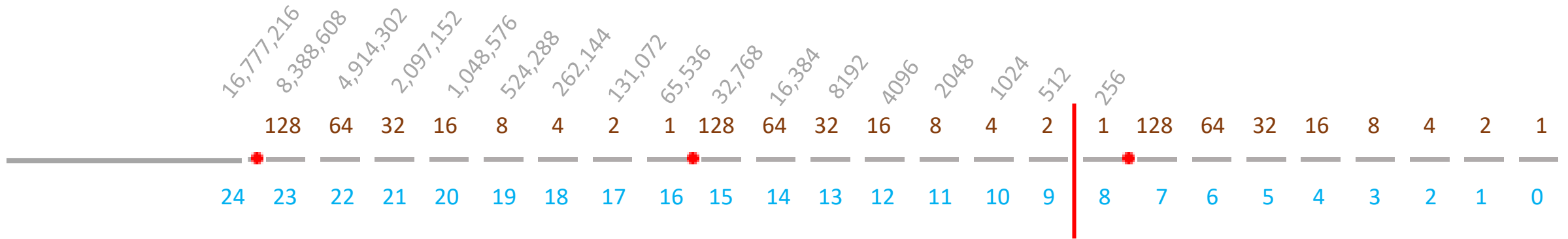
$2^n - 2 = \text{hosts}$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
250	8	1	3 rd	/24	255.255.255.0
2000	11	8	3 rd	/21	255.255.248.0
1000	10	4	3 rd	/22	255.255.252.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

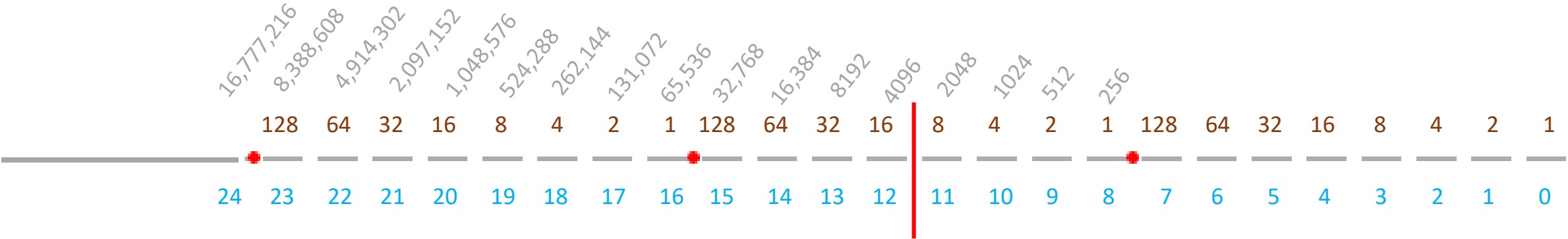
$$2^n - 2 = \text{hosts}$$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
250	8	1	3 rd	/24	255.255.255.0
2000	11	8	3 rd	/21	255.255.248.0
1000	10	4	3 rd	/22	255.255.252.0
500	9	2	3 rd	/23	255.255.254.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

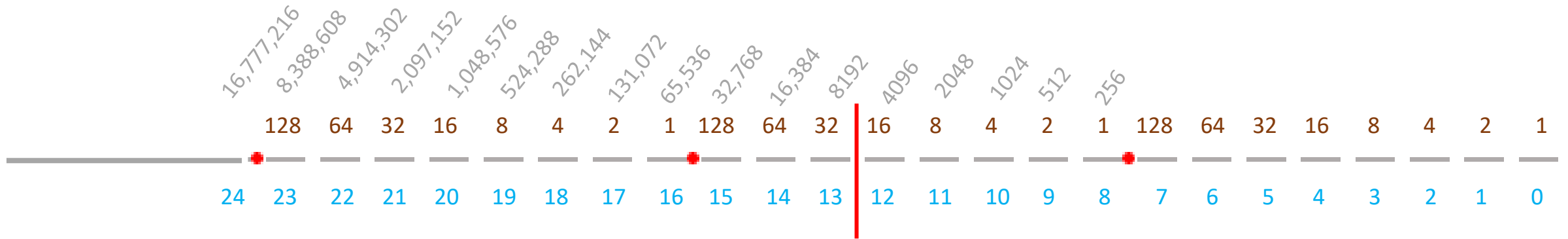
$2^n - 2 = \text{hosts}$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
250	8	1	3 rd	/24	255.255.255.0
2000	11	8	3 rd	/21	255.255.248.0
1000	10	4	3 rd	/22	255.255.252.0
500	9	2	3 rd	/23	255.255.254.0
4000	12	16	3 rd	/20	255.255.240.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

$$2^n - 2 = \text{hosts}$$



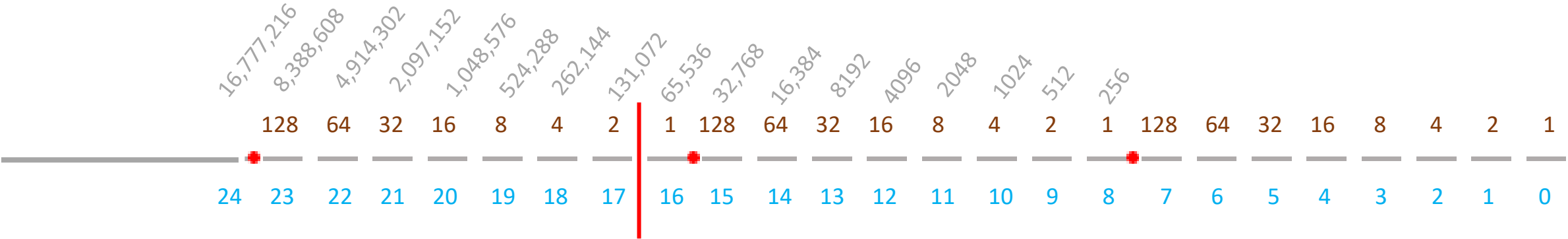
Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
250	8	1	3 rd	/24	255.255.255.0
2000	11	8	3 rd	/21	255.255.248.0
1000	10	4	3 rd	/22	255.255.252.0
500	9	2	3 rd	/23	255.255.254.0
500	9	16	3 rd	/20	255.255.240.0
8000	13	32	3 rd	/19	255.255.224.0

Task 4

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

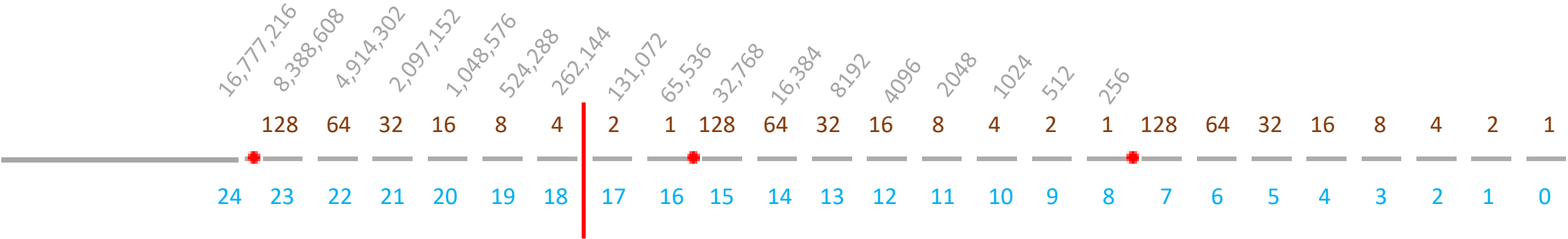
$2^n - 2 = \text{hosts}$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
125,000	17	2	2 nd	/15	255.254.0.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

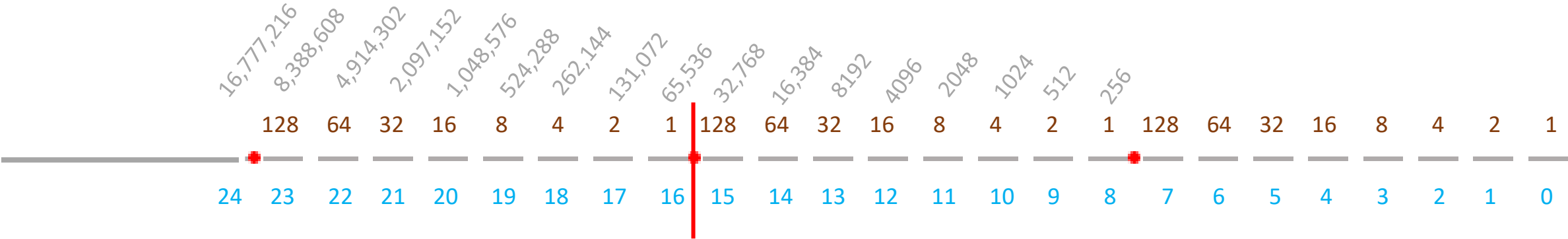
$2^n - 2 = \text{hosts}$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
125,000	17	2	2 nd	/15	255.254.0.0
250,000	18	4	2 nd	/14	255.252.0.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

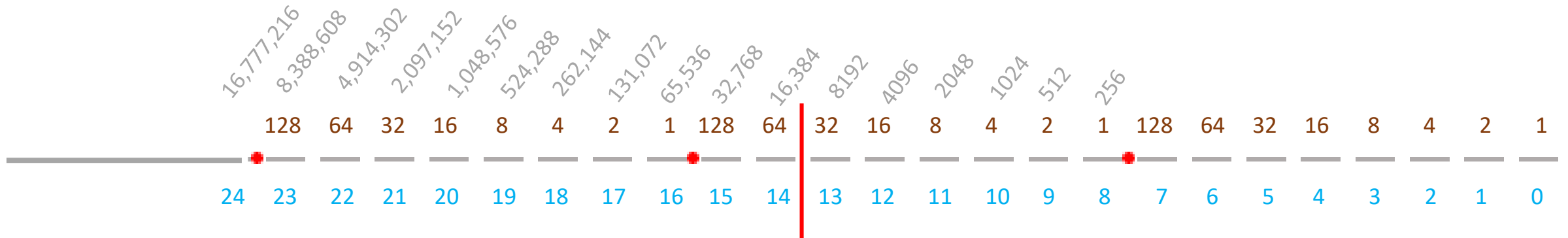
2^n - 2 = hosts



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
125,000	17	2	2 nd	/15	255.254.0.0
250,000	18	4	2 nd	/14	255.252.0.0
65,000	16	1	2 nd	/16	255.255.0.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

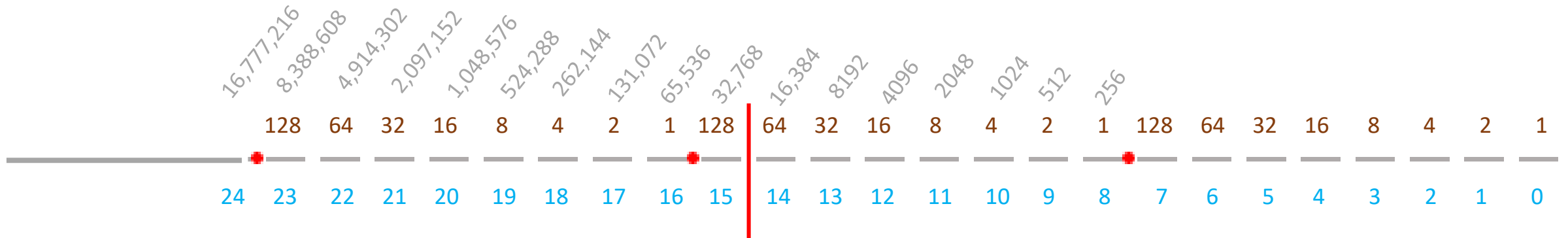
$$2^n - 2 = \text{hosts}$$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
125,000	17	2	2 nd	/15	255.254.0.0
250,000	18	4	2 nd	/14	255.252.0.0
65,000	16	1	2 nd	/16	255.255.0.0
16,000	14	64	3 rd	/18	255.255.192.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

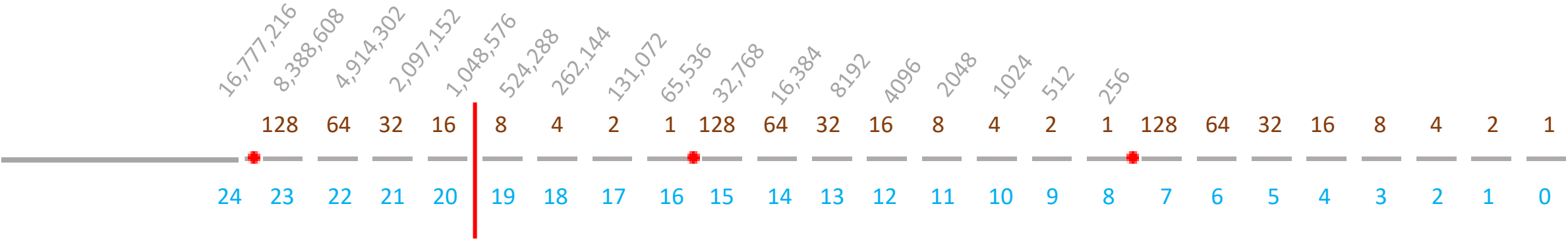
$$2^n - 2 = \text{hosts}$$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
125,000	17	2	2 nd	/15	255.254.0.0
250,000	18	4	2 nd	/14	255.252.0.0
65,000	16	1	2 nd	/16	255.255.0.0
16,000	14	64	3 rd	/18	255.255.192.0
32,000	15	128	3 rd	/17	255.255.128.0

Find the subnet mask that can accommodate the required number of hosts with the least amount of waste

$2^n - 2 = \text{hosts}$



Hosts	Host Bits	Delta	Octet	Mask CIDR	Mask Decimal
125,000	17	2	2 nd	/15	255.254.0.0
250,000	18	4	2 nd	/14	255.252.0.0
65,000	16	1	2 nd	/16	255.255.0.0
16,000	14	64	3 rd	/18	255.255.192.0
32,000	15	128	3 rd	/17	255.255.128.0
1,000,000	20	16	2 nd	/12	255.240.0.0

